

Statistical Analysis for Background Music Data Presented in Lecture Notes no 8

The first experimental design used for the background music data, presented in Lecture Notes no 8, is a Latin square, using the following model

$$Y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk},$$

where $1 \leq i \leq p$, $1 \leq j \leq p$ and $1 \leq k \leq p$. In our experiment $p = 5$. In the model above, the effect of the j th type of background music, which is our treatment factor, is a fixed effect. The other two factors are blocking factors, the day of the week and the week. The levels of the day of the week, α_i are fixed, and the levels of the week, β_k can be fixed or random, depending on how they were chosen. In this experiment, we are only interested in analyzing the effects of the 5 types of background music on the productivity of bank tellers. Therefore, we first test the following null hypothesis:

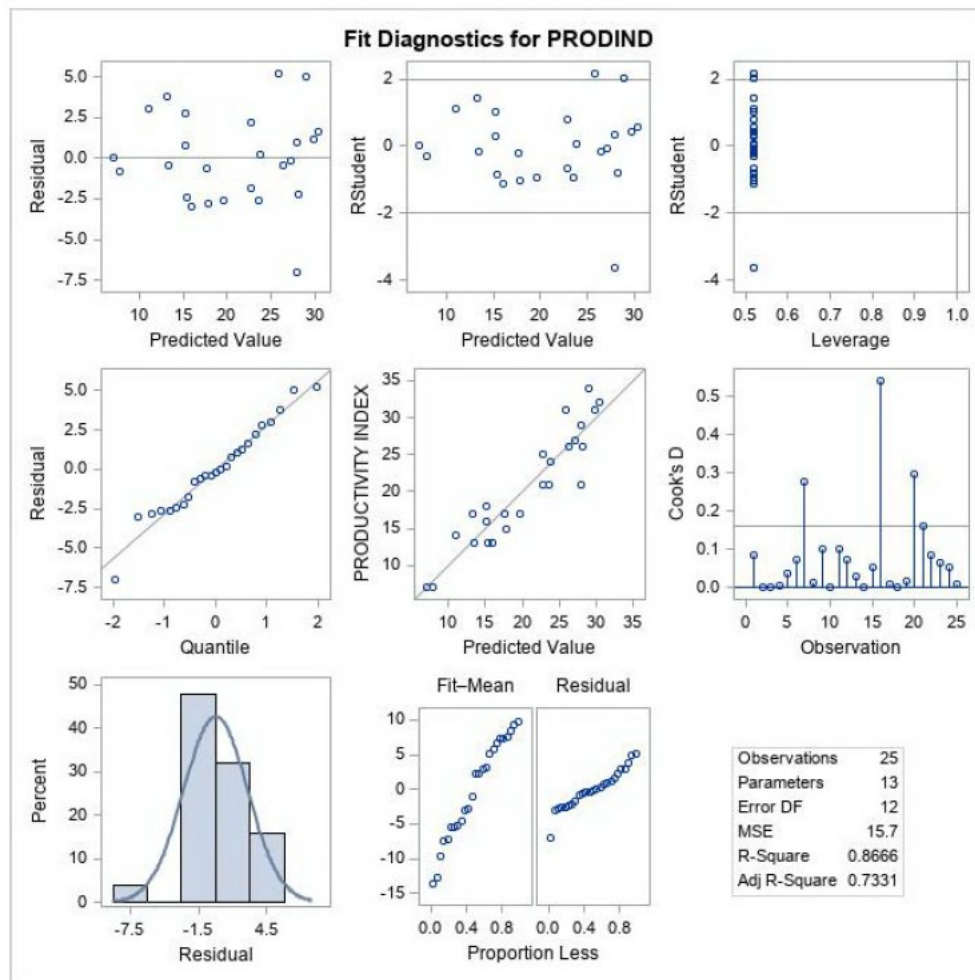
$$H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0$$

vs. the alternative hypothesis

$$H_1: \text{not all the } \tau_j \text{'s are equal to 0.}$$

The SAS file and output for the Music LS design are posted on Husky CT cite for the course. Based on the SAS output, as we mentioned in class, all the underlying assumptions for the model above are valid. For example, the normality assumption for the random errors is confirmed based on the normal probability plot (Residual vs. Quantile plot) in the Fit Diagnostics panel of plots presented below, and the fact that the p-value of the W statistic is equal to .6683, which is larger than .1.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.970913	Pr < W	0.6683



The assumption of linearity of the model and constant variance of random errors is validated via the plot of Residuals vs. Predicted values, as the points in that plot are scattered randomly about the 0 line and do not show any patterns. Based on the plot of the RStudent vs. Predicted Value and Cook's D plot, we can conclude that this data set has no outliers or influential observations.

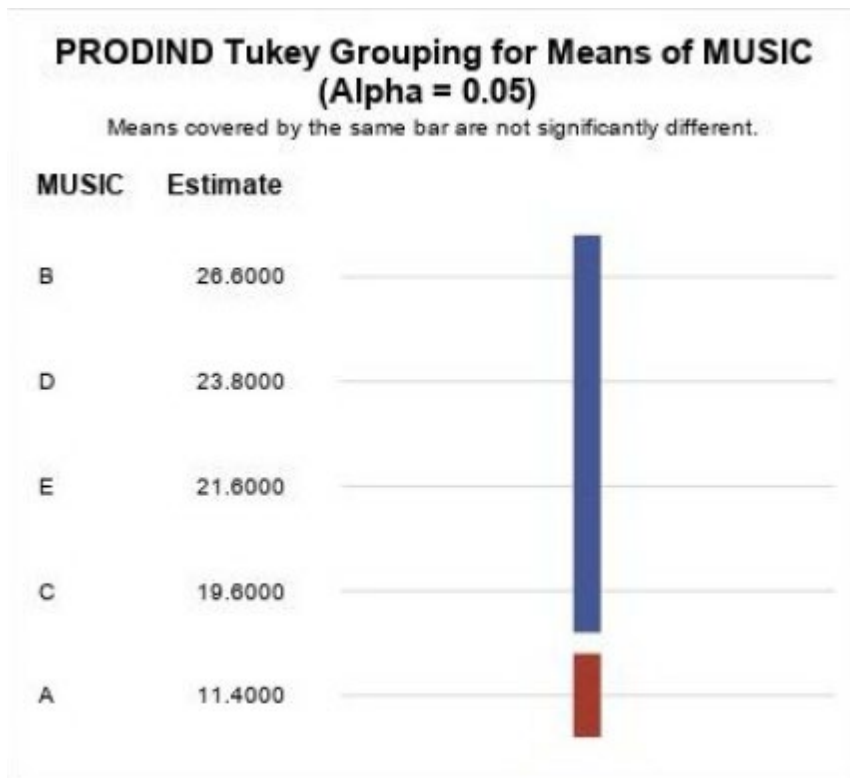
Therefore, from the SAS output it follows from the table below

Source	DF	Type III SS	Mean Square	F Value	Pr > F
MUSIC	4	664.4000000	166.1000000	10.58	0.0007
DAY	4	477.2000000	119.3000000	7.60	0.0027
WEEK	4	82.0000000	20.5000000	1.31	0.3226

that the F statistic for testing the null hypothesis above, has an observed value of 10.58 and a p-value of .0007. Therefore, we can reject this null hypothesis at the .001 level.

This means that some of the types of background music have a significant effect on productivity of bank tellers. To compare the effects of the 5 levels of background music one has to perform Tukey multiple comparisons.

Based on the Tukey multiple comparisons graphical display in the SAS output:



one can conclude that type A music has significant lower productivity index effect than the other four types. Type B, C, D, E are do not have a significant different effect of productivity index.

In the second part of Lecture Notes no 9, we introduced for this data set another blocking variable TIME with 5 levels and re-analyzed the data set as a Graeco-Latin square design, that has the following model:

Model:
$$Y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}$$

where $1 \leq i, j, k, l \leq 5$.

In this model the effects of the five levels of the background music are the treatment levels, denoted by τ_i . We want to test the following hypothesis:

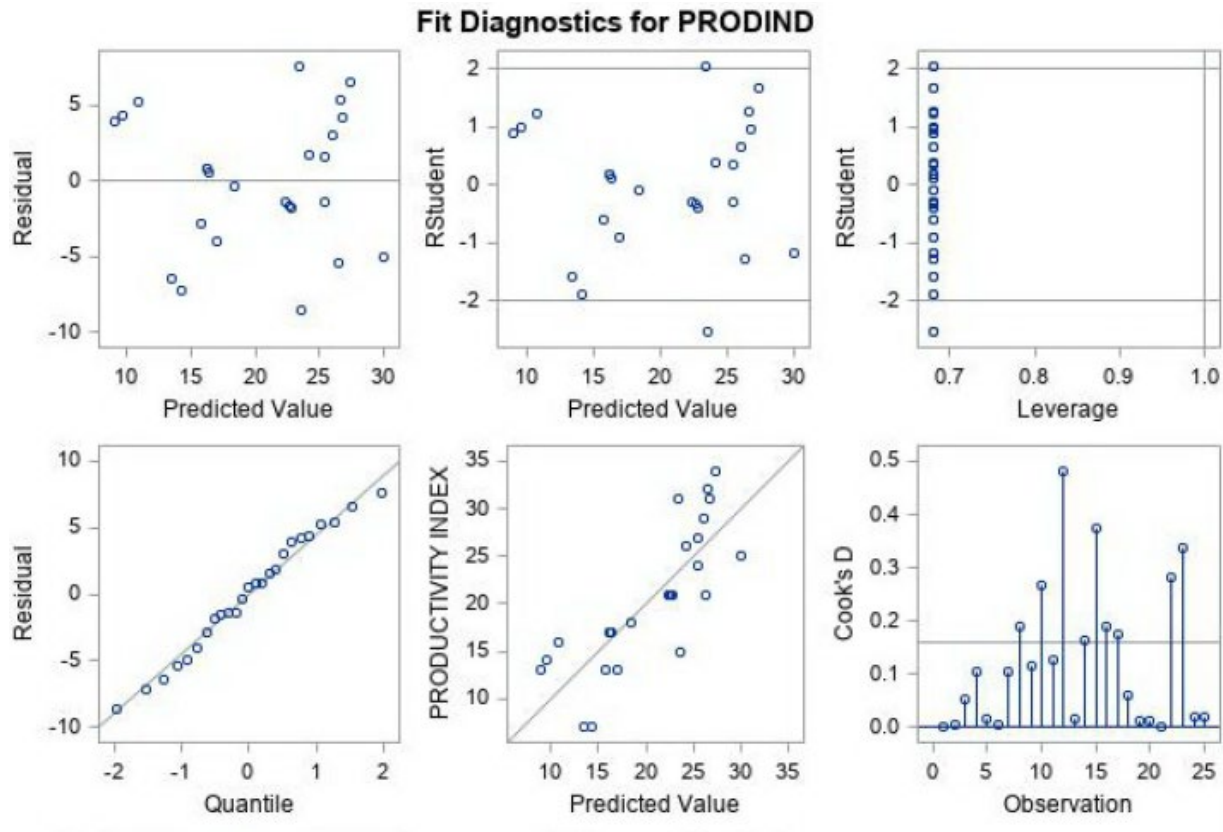
$$H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0$$

vs. the alternative hypothesis

$$H_1: \text{not all the } \tau_j\text{'s are equal to 0.}$$

SAS file and SAS output for Music G_LS design are used to analyze this data modeled by the Graeco-Latin square design. Based on the Fit Diagnostics plots and the p-value of the W statistic presented below we can conclude that all the underlying assumptions for the random errors in our model are valid.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.974473	Pr < W	0.7587



Therefore, if we are testing the null hypothesis above at the .05 level, then based on the Table below we conclude that we accept the null hypothesis, as the F statistic has value 2.73 and a p-value of .1054 > .05. This means that for the Graeco-Latin square design for the music data, the five types of background music have no effect on the productivity of bank tellers.

Source	DF	Type III SS	Mean Square	F Value	Pr > F
MUSIC	4	658.4000000	164.6000000	2.73	0.1054
DAY	4	118.8000000	29.7000000	0.49	0.7416
TIME	4	29.2000000	7.3000000	0.12	0.9709
WEEK	4	94.0000000	23.5000000	0.39	0.8102

Remark: Note that introducing the extra blocking variable TIME made the statistical analysis less sensitive. This might be caused by the low degree of freedom for SS_E .